

Employee Attrition Analysis – Executive Summary

The objective of this project was to identify factors associated with employee attrition and develop a predictive model that can help Human Resources proactively identify employees at risk of leaving the company.

Analysis of 1,470 employee records revealed an overall attrition rate of approximately 16.1%, indicating that most employees remain with the organization while a smaller but significant portion leave. Attrition was notably higher among employees in Sales-related roles and Laboratory Technician positions. Employees who worked overtime, traveled frequently for business, reported lower work-life balance, and experienced longer periods without promotion were also more likely to leave.

Three machine learning models were evaluated, including Logistic Regression, Random Forest, and Gradient Boosting. Logistic Regression provided the strongest overall performance and was selected as the final model due to its predictive ability and ease of interpretation.

The analysis suggests that employee retention efforts should focus on high-risk job roles, reducing excessive overtime, improving work-life balance, and creating clearer career progression opportunities. Early intervention with employees showing multiple risk factors may help reduce turnover and improve workforce stability.

While the model provides valuable guidance, it should be used alongside managerial judgment and employee engagement initiatives because personal and external factors affecting employee decisions are not captured in the available data.